CUSTOM MAGNETICS / SMPS DEVELOPMENT

Custom high-frequency transformers *built for the next bench test.*

Herder Elektronische Systemen develops custom transformer prototypes for switch-mode power supplies, connecting electrical requirements, core selection, winding construction, 3D documentation and measured bench behavior.

DISCUSS A TRANSFORMER PROTOTYPE →

VIEW THE 5 V / 1 A SMPS PROJECT →

FLYBACK SMPS MAGNETICS CUSTOM WINDING 3D MODELING
PROTOTYPE VALIDATION

01 / WHAT CAN BE DEVELOPED

Magnetics shaped *around the converter.*

- Flyback transformer prototypes
- High-frequency isolation transformers
- Custom turns ratios
- Primary, secondary and auxiliary windings
- Gapped ferrite-core prototypes
- RM, EE, EFD, ETD and other practical ferrite geometries where suitable
- Hand-wound engineering samples
- Small prototype quantities
- Revisions of existing transformer designs
- Replacement prototypes where sufficient source information is available

> Feasibility depends on topology, input range, output requirements, switching frequency, isolation needs, core geometry, winding window, thermal limits, safety expectations and component availability.

02 / CUSTOMER INPUTS

Bring the electrical *boundary.*

REQUIREMENTS PANEL

Start with what is known.

Converter topology
Input-voltage range
Output voltage and current
Switching frequency
Controller or primary-side IC
Target power
Number of windings
Auxiliary-bias requirements
Isolation requirement
Core or size constraints
Expected duty cycle
Existing schematic
Existing transformer data
Target quantity

Intended environment
Safety or qualification expectations

> **Do not send sensitive design files through the initial email.**
>
> Secure file exchange can be arranged after the first discussion.

03 / ENGINEERING WORKFLOW

From requirements *to a reviewable prototype.*

01

Define the electrical boundary

Confirm topology, voltage, current, frequency, duty cycle and isolation needs.

02

Select the magnetic structure

Review core material, geometry, effective area, winding window and gap approach.

03

Develop the winding construction

Define turns, wire, winding order, insulation assumptions, polarity and pin assignment.

04

Model and document the assembly

Produce construction drawings, winding-stack visuals and 3D reference models where included.

05

Wind and inspect the prototype

Build the engineering sample and record construction details.

06

Measure and revise

Review inductance, DCR, leakage, switching behavior and converter-level observations where test access permits.

04 / 3D TRANSFORMER MODELING

3D models that make the construction *reviewable.*

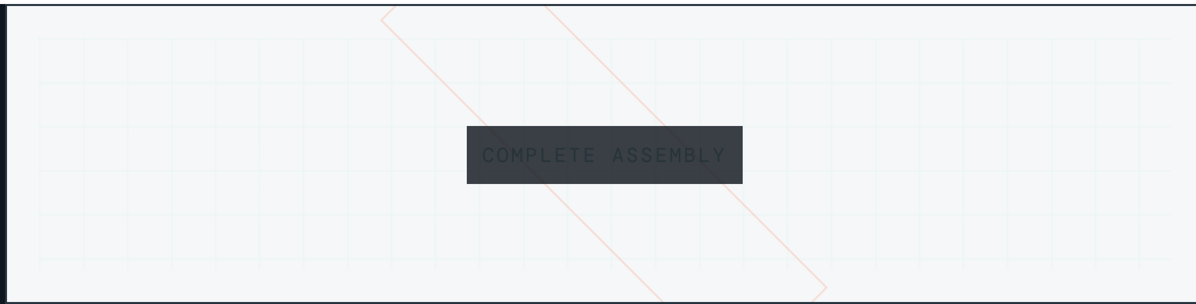
Parametric models can make transformer construction, pin orientation, winding space and mechanical integration easier to inspect before or after the first physical build.

- Parametric modeling of cores, bobbins, pins and assemblies
- Simplified winding-layer models
- Insulation-stack visualization
- Core-gap representation
- Winding-window utilization views
- Exploded assembly views
- Pin numbering and polarity orientation
- PCB-mounted clearance review
- Enclosure and neighboring-component clearance review
- Dimensioned reference drawings
- Technical renders for documentation

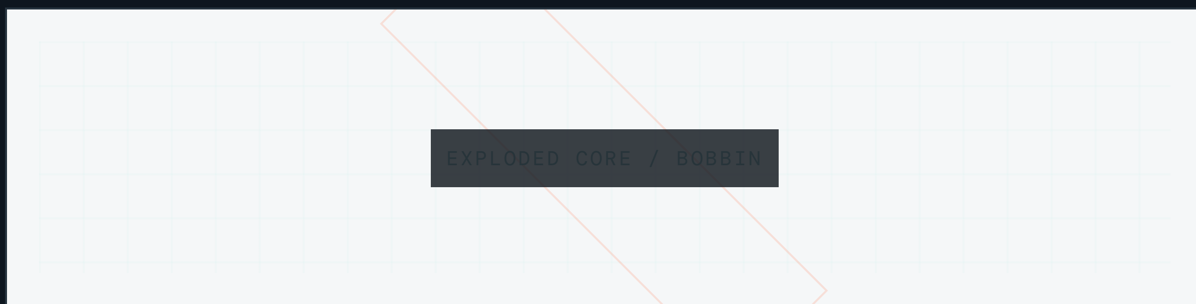
POTENTIAL CAD DELIVERABLES

- Native parametric model
- STEP export
- Dimensioned drawing
- Exploded view
- Winding-stack illustration
- Pinout and polarity diagram
- PCB clearance view

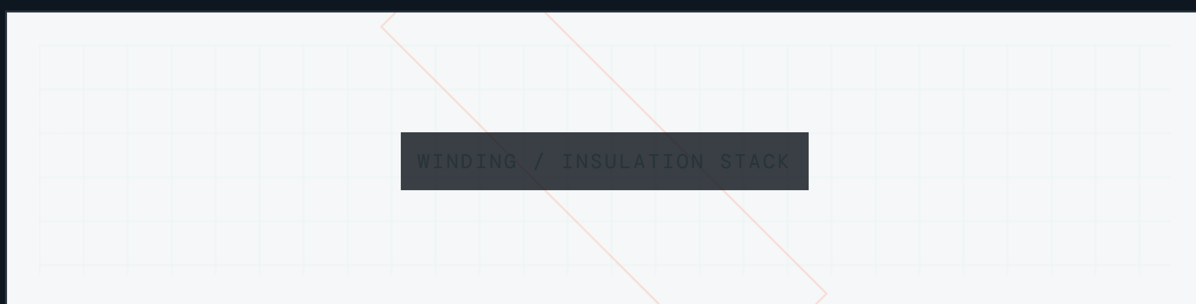
> **The 3D model documents physical construction and integration.**
> It does not by itself validate electromagnetic performance, insulation coordination, thermal behavior, regulatory compliance, manufacturability or production repeatability.



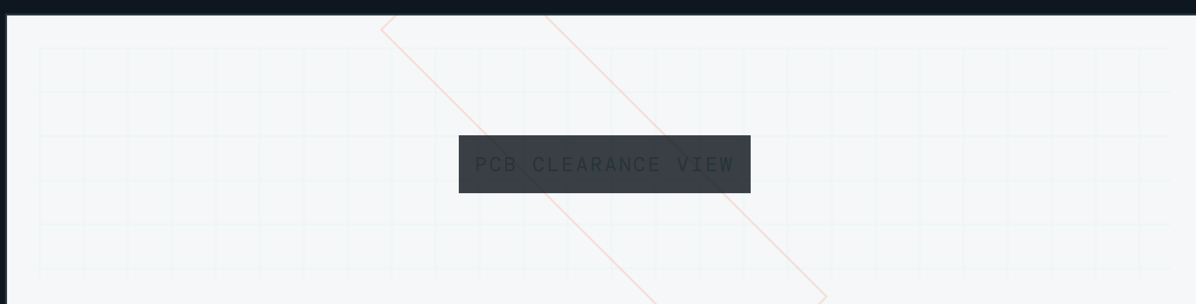
Complete transformer assembly



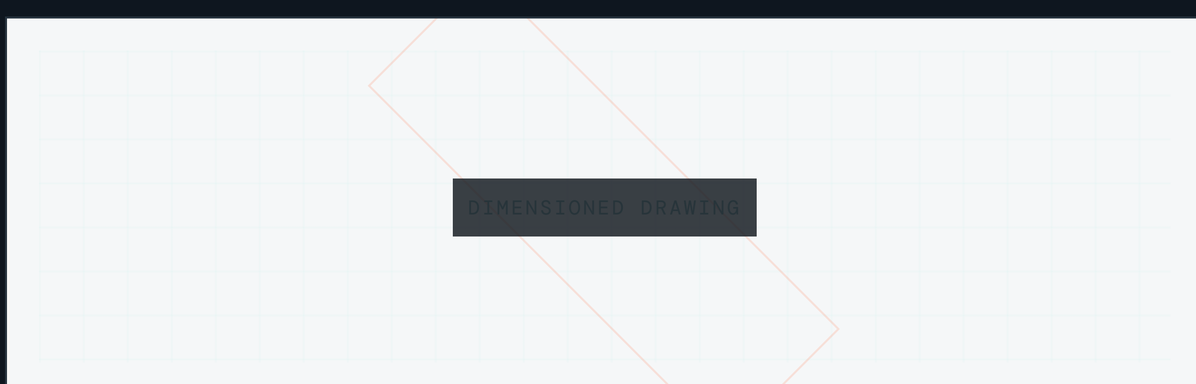
Exploded core and bobbin view



Winding and insulation stack



PCB-mounted clearance view



Dimensioned reference drawing

05 / PROTOTYPE DELIVERABLES

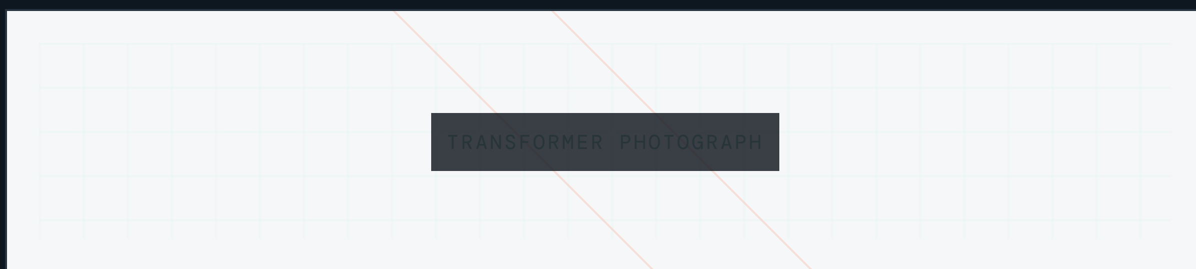
A prototype with enough *information to continue.*

- Wound transformer prototype
- Winding specification
- Turns table
- Wire and insulation notes
- Core and gap information
- Pinout and polarity
- Inductance measurements
- DCR measurements
- Leakage-inductance observations
- Construction notes
- 3D construction model where included
- Bench-test notes
- Revision recommendations

> Exact deliverables depend on the agreed scope, available test access and whether the request covers transformer construction only or converter-level evaluation.

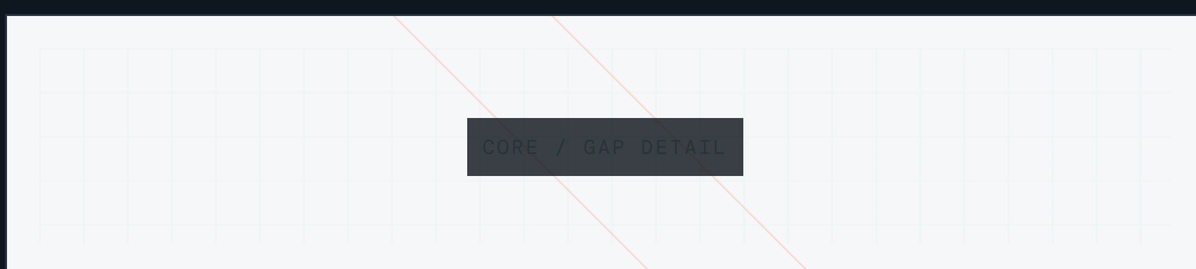
06 / EVIDENCE AND VALIDATION

Measurements with the *test conditions attached.*



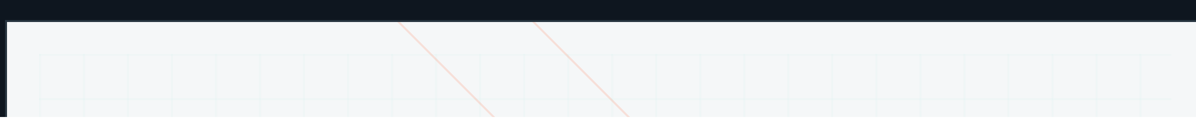
Caption / provenance / revision

Turns, polarity and pin assignment



Core geometry and gap detail

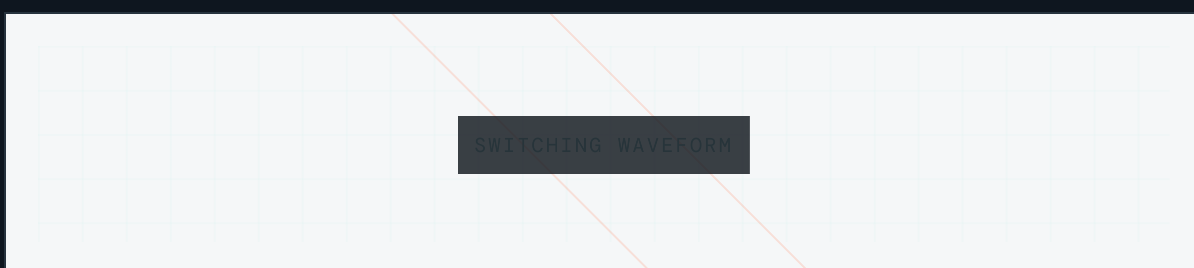
Transformer in converter context



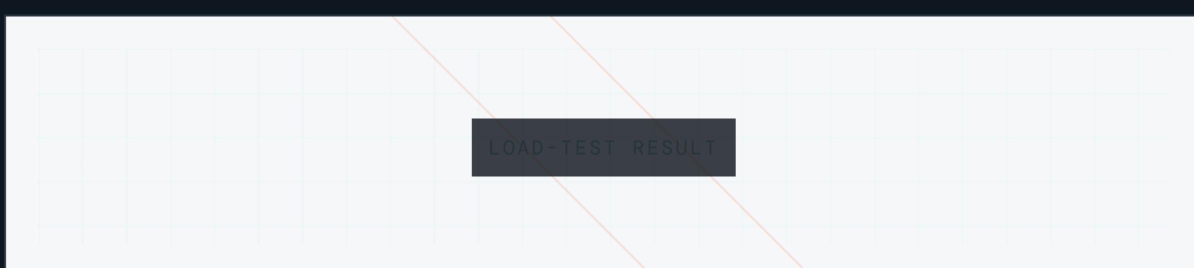
Instrument / date / test condition

Instrument / date / winding

Winding condition and method



Operating point / probe / revision



Input, load and thermal context

Model revision and CAD provenance

Every evidence block should identify source or provenance, instrument, test date, operating condition and prototype revision. No values, plots, dimensions, photographs or validation claims are invented here.

> **Prototype magnetics are engineering samples.**
>
> Suitability for production, reinforced insulation, regulatory compliance, long-term reliability, thermal performance and manufacturing repeatability requires separate analysis, testing and qualification.

07 / RELATED WORK

See the transformer *in converter context.*

PUBLIC TECHNICAL PROTOTYPE

5 V / 1 A SMPS

Offline flyback prototype with custom transformer work, schematics, waveform captures and bench-validation notes.

VIEW PROJECT →

VIEW REPOSITORY ↗

POWER ELECTRONICS PROTOTYPING →

START A TECHNICAL DISCUSSION

Bring the *converter requirements.*

Share the topology, electrical targets, switching frequency, intended quantity and what has already been measured. Secure file exchange can follow the initial discussion.

DISCUSS A TRANSFORMER PROTOTYPE →

Or write directly: info@herdersystemen.com

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